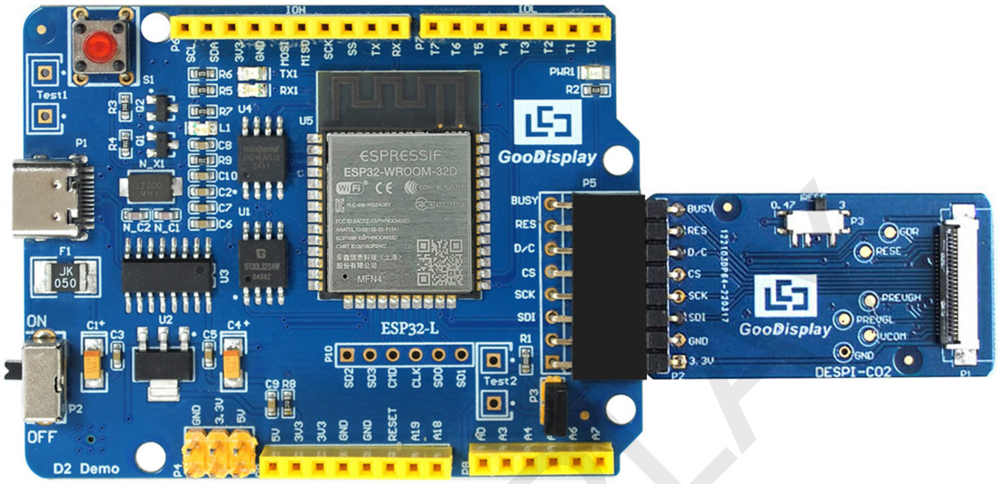
E-Paper Display Evalution Kit for ESP32

ESP32-L(C02)

GooDisplay

Dalian Good Display Co., Ltd.

Product Specifications



Customer	**Standard**
Description	Evaluation Kit For E-paper Display
Model Name	ESP**32**-L(C02)
Date	2022/**07**/07
Revision	**2.0**

	Design Engineering		
	Approval	**Check**	**Design**
	刘玉宝印	修凤燕印	吴良之印

Zhongnan Building, No.18, Zhonghua West ST,Ganjingzi DST,Dalian,CHINA
Tel: +86-411-84619565
Email: info@good-display.com
Website: www.good-display.com

Contents

1.Overview

ESP32-L evaluation kit is used to help users develop e-paper display projects with provided source code to create more differentiated solutions. It is designed for SPI e -paper display. It supports driving Good Display's black-white e-paper display and three-color (black, white and red/Yellow) e-paper display: 0.97'', 1.54'',2.13'', 2.66'' , 2.7'', 2.9'', 3.71'', 4.2'', 5.83'' and 7.5''. And it is added the functions of USB serial port and LED indicator light, Reset button, font chip, Flash chip and etc.

ESP32-L (C02) development kit consists of motherboard ESP32-L for EPD and connector board DESPI-C02.
Esp32-L (C02) development kit is only for users to develop and drive electronic paper display screens. The application of WiFi, Bluetooth and other functions requires users to develop on their own according to the project.

2. Structure Specification

Parameter	Specification
Model	ESP32-L（C02）
Platform	Arduino
Dimension	Mother Board：70mm x 54mm (ESP32-L) Adapter：41mm x 22mm (DESPI-C02)
Power Interface	Type-C
Example Code	Available
Operating Temp.	-20 ℃ ~ 70 ℃
Main Function	Learn to drive E-paper display; Test and evaluate e-paper display; Support secondary development
Additional Function	Type-C,LED indicator, Reset button, Font chip, Flash chip, Current detection

3. Functions

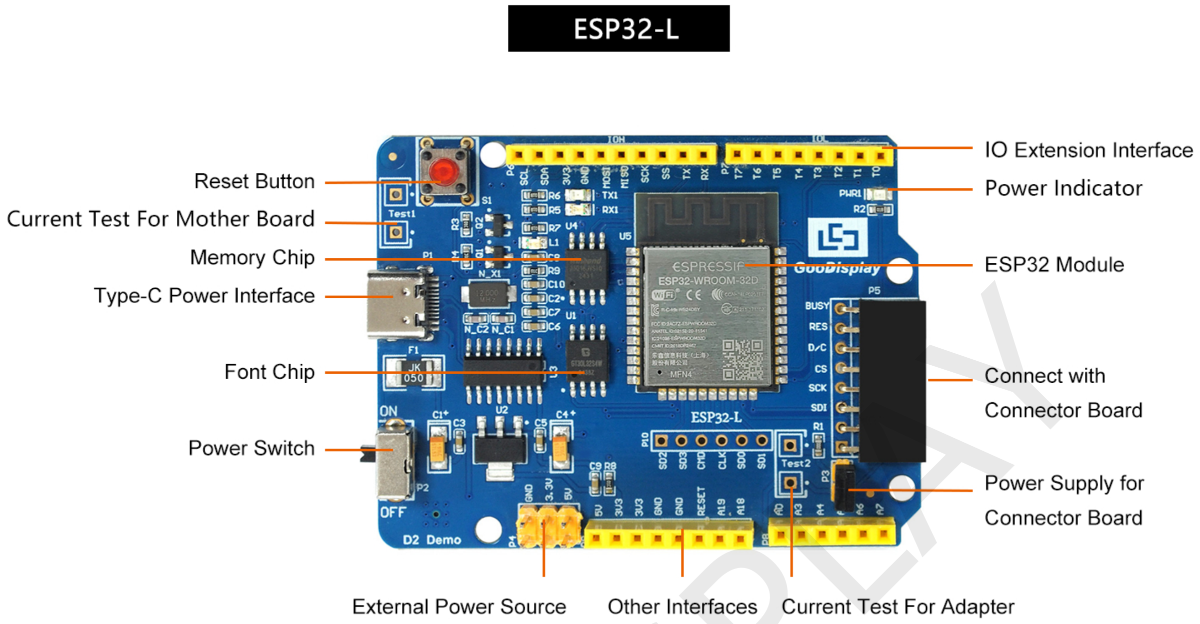


Figure 1 : ESP32-L

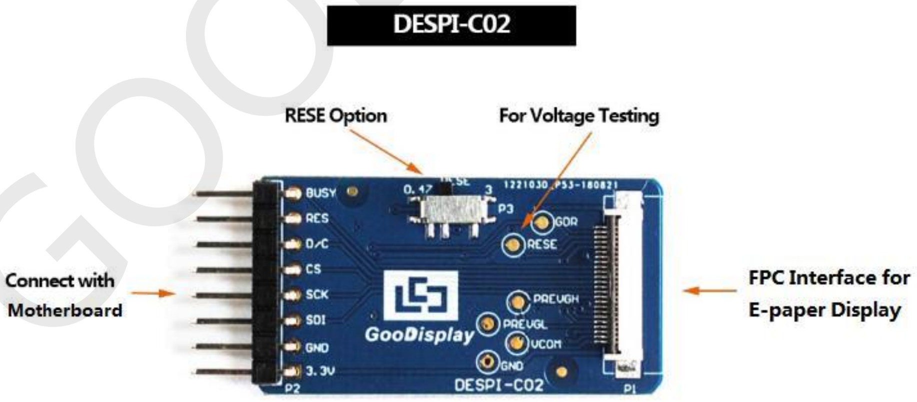


Figure 2 : DESPI-C02

3.1 Power Supply

The input voltage of this board is DC5V, which is powered by the USB port

3.2 USB to serial port transmission

This development board uses USB to serial port communication. Users should install CH340 driver on computer before downloading program.

3.3 P3 short-circuit jumper

P3 short-circuit jumper controls DESPI-C02's power supply, which is e-paper's power supply. Be sure to short it when using.

3.4 Current measurement

The development kit supports current measurement of motherboard ESP32-L and DESPI-C02.

1) motherboard ESP32-L: Power off and make series connection between ampere meter and TEST1.
2) DESPI-C02: Power on and take off the short-circuit jumper P3, then make series connection between ampere meter and TEST2. Put on the short-circuit jumper P3 after measurement.

3.5 I/O port extension

This development board led out the digital I/O 0~13 and the analog I/O 0~5 for development.

3.6 LED indicator light

There is a indicator light reserved for developing.

3.7 Reset key

This development board contains a reset key for users operation.

3.8 Expanded Functions

Built-in Chinese font chip GT30L32S4W.
Built-in data storage chip W25Q16.

4. Connection Mode and RESE Selection

4.1 Connection between e-paper and development board

Connect DESPI-C02 to ESP32-L for EPD as shown in Figure 3. Connect e-paper FPC to DESPI-C02 as shown in Figure 4. (Please note the direction of the e-paper.)

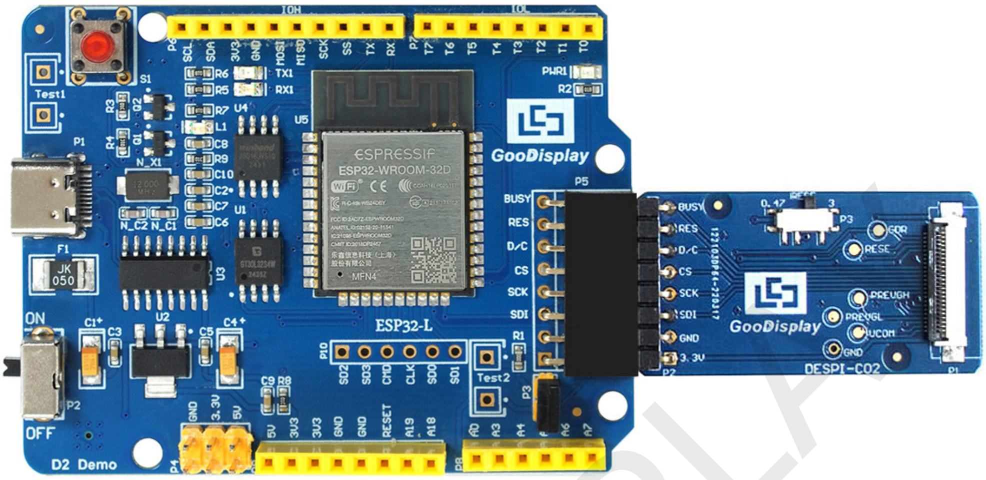


Figure 3 : Connection between ESP32-L for EPD and DESPI-C02

4.2 Connection between e-paper and adapter board

1) Identify the front and back sides of the e-paper, and insert the e-paper into the adapter plate with the front side facing up

Note: The mirror side of the e-paper is facing down and the display side is facing up.

Figure 4 Schematic diagram of e-paper and adapter board plugging

2) Insert the e-paper FPC gold finger upward into the P1 connector of the adapter board as shown in Figure 5.

(1) FPC connector	
(2) Open the connector	
(3) Insert the FPC	
(4) Close the connector	

Figure 5 : Connection between DESPI-C02 and e-paper

4.3 Selection of RESE Resistor for Adapter Board

The DIP switch on the adapter board is used to select the RESE resistor. Different models of e-paper require matching RESE resistors, and selecting the wrong resistor may result in the e-paper failing to refresh properly.

Note: When designing actual products, users must strictly follow the circuit design specified in the e-paper product specification.

- 0.47Ω is suitable for UC (Jinghong) series IC e-paper, including UC8151D, UC 8276, UC8179, etc.
- 3Ω is suitable for SSD (Solomon) series IC e-paper and all types of four-color e-paper.

5. Program Downloading

This development board uses a serial port for program downloading and requires the Arduino IDE, a Type-C data cable, and the CH340 driver. The operation steps are as follows:

5.1 Installing Arduino IDE and ESP32 Development Board

1) Download the Arduino IDE software from the official Arduino website: www.arduino.cc. (Using version V2.3.3 as an example.)
2) In Arduino IDE, go to Tools > Board > Boards Manager.
3) In Boards Manager, search for ESP32, select the appropriate version, and click INSTALL.

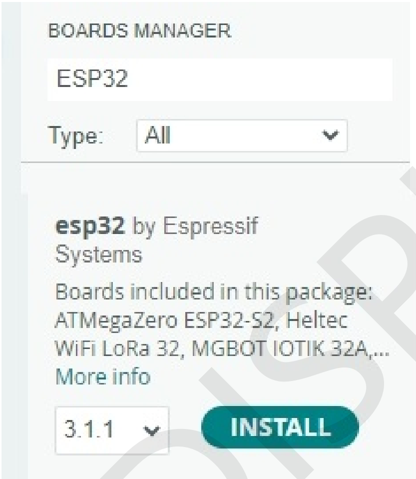


Figure 6: ESP32 Development Board Library Installation Diagram

4) After the ESP32 development board installation is complete, you will see the relevant board information in Tools > Board.

5.2 Downloading the ESP32 Development Board Driver

1) Connect the USB interface of the development board to the computer using a Type-C data cable.
2) Open the ino project file in the e-paper driver program folder using Arduino IDE, as shown in the figure below.

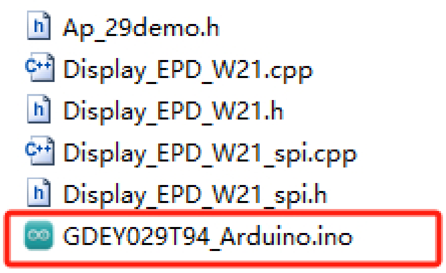


Figure 7: Opening the ino Project File

3) Steps to Download the E-paper Driver Program for the ESP32 Development Board

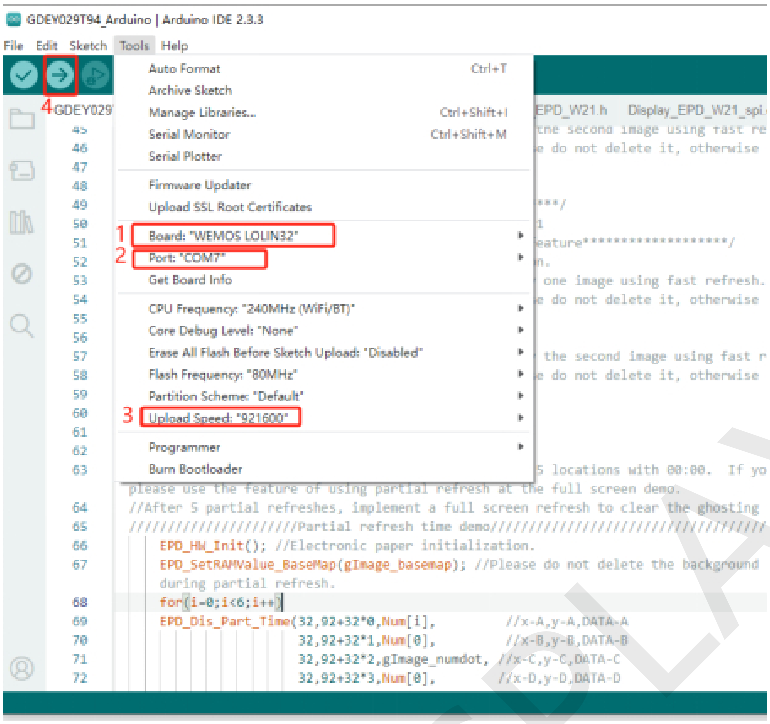


Figure 8: Program Download Operation Steps

1--Select the development board model "WEMOS LOLIN32".
2--Select the COM port.
3--Select the serial baud rate "921600".
4--Download the program to the development board.

4) After a successful download, first power off the development board, connect the e-paper display to the adapter board, then power on again. The e-paper will display the image normally.

5.3 Using the ESP32epdx E-paper Official Library

Users can search for "ESP32epdx" inManage Librariesin Arduino IDE and install this official e-paper library. This library includes common Good Display e-paper drivers, supports GUI functionality, and will continuously add new e-paper application features, such as controlling the e-paper display via Bluetooth, WiFi, etc.

Github: https://github.com/gooddisplayshare/ESP32epdx

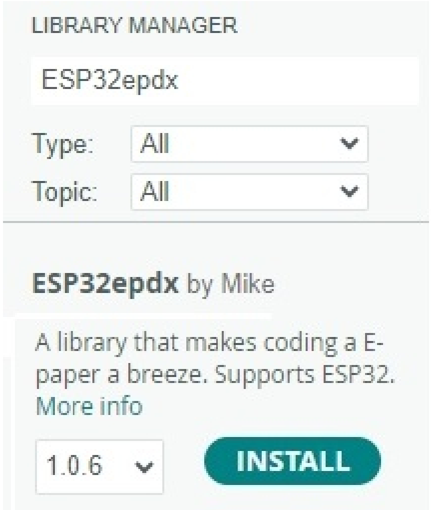


Figure 9: ESP32epdx Official Library Installation Diagram